PUNEET.

Department of Mathematics, IIT Madras

Scientific Computing Advanced Programming Lab (MA-5105)

QUIZ-III

21 October, 2026

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- This question paper has two questions, totalling to 15 marks and all are to be answered.
- As announced, malpractice can result in disciplinary actions as per the Senate mandate.

Q. 1 Let $f : \mathbb{R}^n \to \mathbb{R}$ be a differentiable function that we wish to minimize. The steepest descent method generates a sequence of iterates according to

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha \nabla f(\mathbf{x}_k),$$

where $\alpha > 0$ is a sufficiently small, fixed step size.

Write the pseudocode for the steepest descent algorithm to find a local minimizer of f. Your algorithm should:

(a) Initialize the starting point $\mathbf{x}_0$, tolerance $\varepsilon > 0$, step size $\alpha > 0$, and maximum number of iterations.

(b) Compute the gradient $\nabla f(\mathbf{x}_k)$ at each iteration.

(c) Check convergence using the stopping criterion

$$\|\nabla f(\mathbf{x}_k)\|_2 < \varepsilon.$$

(d) Update the iterate using the steepest descent formula given above.

(e) Terminate when the stopping criterion is satisfied or the maximum number of iterations is reached.

[5 Marks]

Q. 2 Consider the constrained optimization problem

$$\min_{(x,y)\in\mathbb{R}^2} f(x, y) = (x - 2)^2 + (y - 1)^2$$

subject to the constraint

$$x + y = 2.$$

The penalty method converts this constrained problem into an unconstrained optimization problem by introducing the penalized objective function

$$F_\mu(x, y) = f(x, y) + \frac{\mu}{2}(x + y - 2)^2,$$

where $\mu > 0$ is the penalty parameter.

Write the pseudocode for the quadratic penalty method to solve the given constrained optimization problem. Your algorithm should:

(a) Initialize (x_0, y_0), the penalty parameter $\mu > 0$, a tolerance $\varepsilon > 0$, and a maximum number of outer iterations.

(b) Construct the penalized objective function $F_\mu(x, y)$.

(c) Minimize F_μ using the steepest descent method with a sufficiently small, fixed step size.

(d) After obtaining an approximate minimizer, check whether the constraint violation satisfies

$$|x + y - 2| < \varepsilon.$$

(e) If the constraint violation is too large, increase the penalty parameter by setting

$$\mu \leftarrow 10\mu,$$

and repeat the minimization. Terminate when the constraint tolerance is satisfied or the maximum number of outer iterations is reached.

(f) Justify the penalization term i.e. discuss the role of μ and the term multiplied to it. [4 Marks]

[6 + 4 Marks]